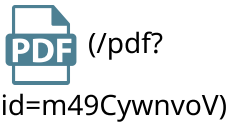




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Full-Season Agent Evaluation in Soybean Farm Operations under Real-World Agricultural Process Dynamics



ICML 2026 Workshop AIWILD Submission310 Authors

📅 09 May 2026 (modified: 09 May 2026) 📁 ICML 2026 Workshop AIWILD Submission
👁️ AIWILD, Reviewers, Authors 📄 Revisions (/revisions?id=m49CywnvoV)
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Keywords: Agents in the Wild, Long-Horizon Evaluation, Agricultural AI, Tool Use, Process Models

Abstract:

In this preliminary field study, we evaluate contemporary agent systems on the operational pipeline of an industry-grade soybean research farm. The environment is not a bespoke benchmark wrapper around an LLM tool interface; it is built from the physics-grounded process models, sensing infrastructure, machinery workflows, and seasonal historical operation records used to manage and study the farm. We evaluate nine representative agent-controller methodologies across three levels of farm operations, ranging from atomic tasks and episode chains to full-season scenarios that span the entire harvest cycle. Our results show that human-expert context is the dominant factor in long-horizon performance: without it, full-season agents remain 34% below the human-oracle soybean yield, while expert context reduces this shortfall to 7%. We observe that full-path agentic correctness and crop yield drop under longer horizons, and that agriculture-domain LLM-as-an-Expert guidance can recover yield while still lagging human experts. With these field observations from an operating research farm, we hope to inform AI research in domains where actions interact with physical processes, delayed feedback, and production-level outcomes.

Track: Short Paper (4 pages)

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This paper evaluates how well different AI agent architectures can manage a soybean farm using a realistic digital twin environment. The authors test the agents on short tasks, multi

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step episodes, and full season farming scenarios. They found that agents do great on short tasks but struggle with long term planning. They also discovered that human expert advice is still much better than LLM generated advice, and that using multiple agents actually hurts performance due to coordination issues.

Official Review by Reviewer T7cj (👁 Vivek Kumar Mishra (/profile?id=~Vivek_Kumar_Mishra1))

📅 13 May 2026, 10:49 👁 Program Chairs, Reviewer T7cj

Review:

Quality The quality of the experiment is excellent. Using a digital twin based on real farm sensors and crop models is a very smart way to test agents in a complex environment with delayed feedback. The metrics they use, like temporal correctness and actual crop yield shortfall, are grounded in reality and make the results highly credible.

Clarity The paper is very easy to read. The authors clearly explain the farm setup, the agent models they tested, and the three levels of task complexity. The results are presented in a way that is easy to digest, even for someone who is not an expert in agriculture.

Originality Testing agents on full season agricultural management is a really fresh idea. While there are plenty of agent benchmarks out there, applying them to a physical process with delayed consequences like farming is a great contribution. The finding that multi agent systems actually perform worse in this setting is also a surprising and original insight.

Significance This is a highly significant paper for the AI agents community. It highlights the gap between doing quick software tasks and managing long term physical processes. The insights about the limits of multi agent coordination and the ongoing need for human context will be very useful for researchers building agents for the real world.

Pros and Cons Pros:

Very realistic testing environment using a digital twin of a real farm. Tests long term consequences instead of just immediate pass or fail tasks. Uncovers surprising findings about the limitations of multi agent setups. Cons:

It is still a simulation, so real physical robotics failures are not fully captured. The study only looks at soybeans, so it is not clear how well these findings apply to other types of farming. I would have liked to see more detail on exactly why the multi agent systems failed to coordinate so poorly.

Rating: 7: Good paper, accept

Confidence: 3: The reviewer is fairly confident that the evaluation is correct

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